

ENGR 225: Mechanics 1

Homework Coversheet

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DATE: _____

ASSIGNMENT: Homework 3

Level of involvement of AI, LLMs, or related tools in the completion of this assignment:

☐ None

☒ Minor

☐ Major

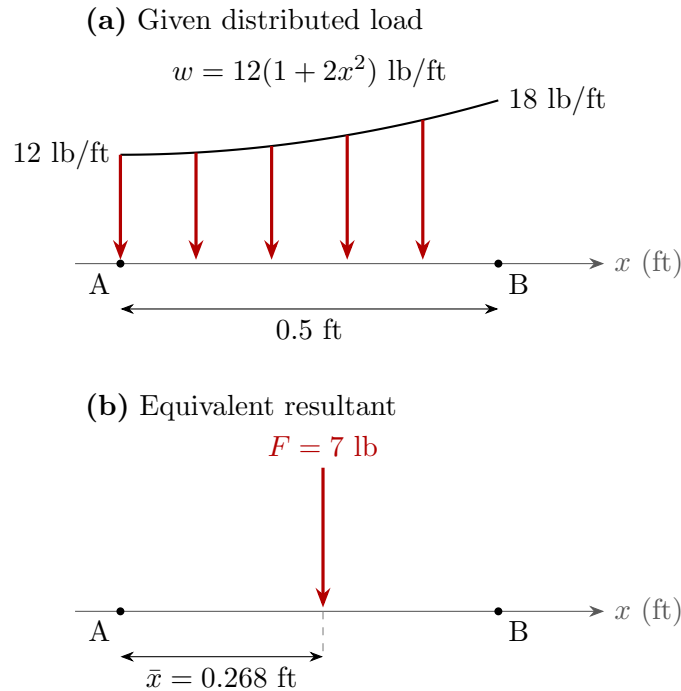
☐ Extensive

Mark the box that applies (e.g., replace ☐ with ☒, or type an X beside it, before compiling/printing). Your disclosure does not affect your grade – it keeps you accountable for the work you are doing.

Problem 1

Given / Find: A seat-restraint pressure on the head varies parabolically, $w(x) = 12(1 + 2x^2)$ lb/ft, for $0 \leq x \leq 0.5$ ft measured from A (12 lb/ft at A, 18 lb/ft at B). **Find** the equivalent resultant force F and its location $\bar{x}$ from A.

Sketch:



Equations & Solution:

The resultant force equals the area under the load curve, and it acts at the centroid of that area:

$$\begin{aligned}
 F &= \int_0^{0.5} w \, dx = \int_0^{0.5} 12(1 + 2x^2) \, dx = 12 \left[x + \frac{2}{3}x^3 \right]_0^{0.5 \text{ ft}} \frac{\text{lb}}{\text{ft}} \\
 &= 12 \left(0.5 + \frac{2}{3}(0.125) \right) \text{ lb} = 12(0.5833) \text{ lb} = 7.00 \text{ lb} \\
 \bar{x} &= \frac{\int_0^{0.5} x w \, dx}{\int_0^{0.5} w \, dx} = \frac{12 \left[\frac{x^2}{2} + \frac{x^4}{2} \right]_0^{0.5} \text{ lb}}{7.00 \text{ lb}} = \frac{12(0.125 + 0.03125) \text{ lb}\cdot\text{ft}}{7.00 \text{ lb}} = \frac{1.875 \text{ lb}\cdot\text{ft}}{7.00 \text{ lb}} = 0.268 \text{ ft}
 \end{aligned}$$

Check: $\bar{x} = 0.268$ ft is slightly greater than 0.25 ft (the midpoint) because the load is heavier at B.

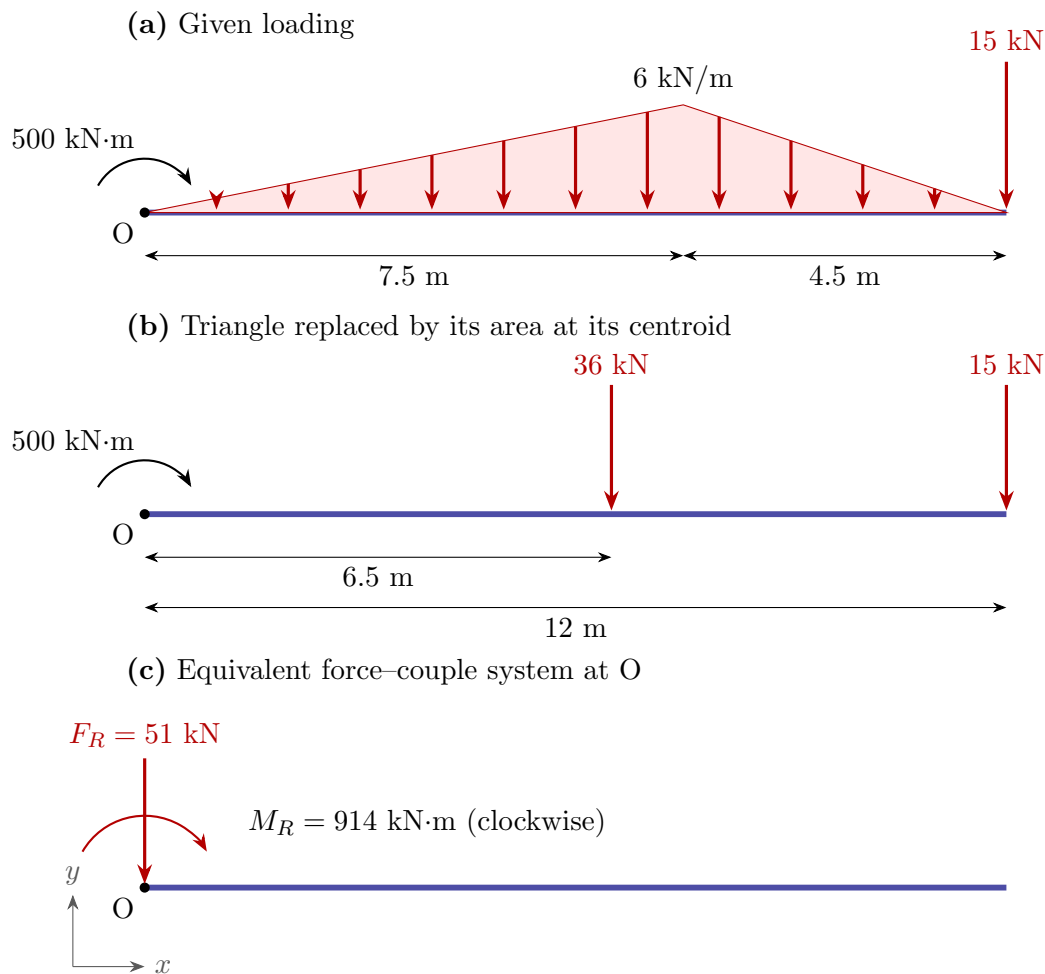
Answer

$F = 7.00$ lb (in the direction of the pressure), acting at $\bar{x} = 0.268$ ft from A.

Problem 2

Given / Find: A 12 m beam carries a triangular load (peak 6 kN/m at $x = 7.5$ m, zero at both ends), a 15 kN point load at the right end ($x = 12$ m), and a 500 kN·m clockwise couple at O. **Find** the equivalent single force F_R and couple moment M_R acting at O. (Counterclockwise moments are taken positive.)

Sketch:



Equations & Solution:

Step 1 – replace the triangular load by its area, placed at its centroid. The triangle has base 12 m and height 6 kN/m; its centroid lies one-third of the sum of the vertex positions (0, 7.5, 12 m) from O:

$$W_{\Delta} = \frac{1}{2}(12 \text{ m})(6 \text{ kN/m}) = 36.0 \text{ kN}, \quad \bar{x}_{\Delta} = \frac{0 + 7.5 + 12}{3} \text{ m} = 6.5 \text{ m}$$

Step 2 – resultant force (upward positive):

$$\sum F_y: \quad F_R = -36.0 \text{ kN} - 15.0 \text{ kN} = -51.0 \text{ kN}$$

Step 3 – resultant moment about O:

$$\begin{aligned} M_R = \sum M_O &= -500 \text{ kN}\cdot\text{m} - (36.0 \text{ kN})(6.5 \text{ m}) - (15.0 \text{ kN})(12 \text{ m}) \\ &= -500 \text{ kN}\cdot\text{m} - 234 \text{ kN}\cdot\text{m} - 180 \text{ kN}\cdot\text{m} = -914 \text{ kN}\cdot\text{m} \end{aligned}$$

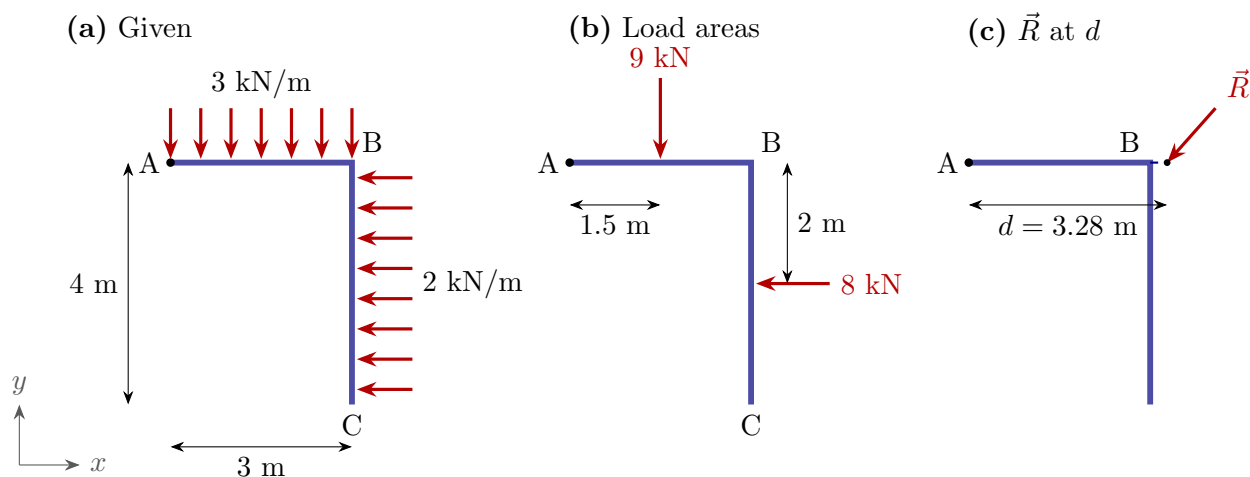
The negative signs mean the resultant force points downward and the couple is clockwise.

Answer
$F_R = -51.0 \text{ kN}$ (51.0 kN downward) and $M_R = -914 \text{ kN}\cdot\text{m}$ (914 kN·m clockwise), both acting at O.

Problem 3

Given / Find: Frame ABC carries a 3 kN/m load downward along the 3 m horizontal member AB and a 2 kN/m load acting leftward along the 4 m vertical member BC. **Find** the single equivalent resultant force $\vec{R}$ and the distance d from A to where its line of action crosses the horizontal line through AB.

Sketch:



Equations & Solution:

Step 1 – load areas and their locations (from A; x right, y up):

$$W_1 = (3 \text{ kN/m})(3 \text{ m}) = 9 \text{ kN downward at } (x, y) = (1.5 \text{ m}, 0)$$

$$W_2 = (2 \text{ kN/m})(4 \text{ m}) = 8 \text{ kN leftward at } (x, y) = (3 \text{ m}, -2 \text{ m})$$

Step 2 – resultant force:

$$\vec{R} = \sum \vec{F} = (-8 \hat{i} - 9 \hat{j}) \text{ kN}, \quad |\vec{R}| = \sqrt{8^2 + 9^2} = 12.0 \text{ kN}$$

Step 3 – moment about A (CCW positive), using $M = xF_y - yF_x$ for each load:

$$M_A = \underbrace{(1.5 \text{ m})(-9 \text{ kN}) - 0}_{W_1} + \underbrace{(3 \text{ m})(0) - (-2 \text{ m})(-8 \text{ kN})}_{W_2} = -13.5 \text{ kN}\cdot\text{m} - 16.0 \text{ kN}\cdot\text{m} = -29.5 \text{ kN}\cdot\text{m}$$

Step 4 – position of the line of action. If $\vec{R}$ acts at $(d, 0)$ on the line through AB, only R_y produces a moment about A (R_x passes through A's horizontal line, so its arm is zero):

$$M_A = d R_y \Rightarrow -29.5 \text{ kN}\cdot\text{m} = d (-9 \text{ kN}) \Rightarrow d = \frac{29.5}{9} \text{ m} = 3.28 \text{ m}$$

Since $d > 3 \text{ m}$, the line of action crosses the horizontal line through AB slightly to the right of B.

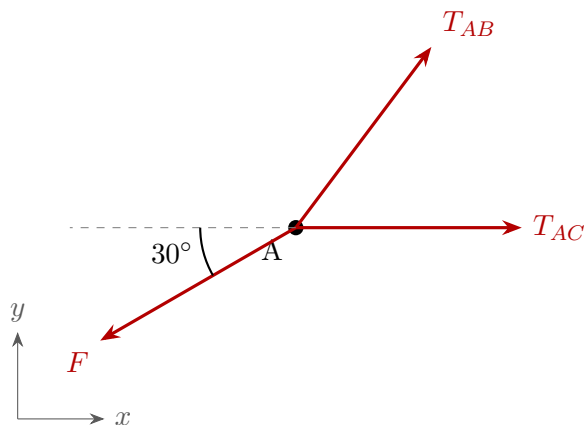
Answer

$\vec{R} = -8.00\hat{i} - 9.00\hat{j}$ kN (12.0 kN total), acting at $d = 3.28$ m from A; the equivalent couple moment is zero.

Problem 4

Given / Find: Chains AB (3-4-5 slope, rising to the right) and AC (horizontal) hold joint A, and a force $\vec{F}$ acts down and to the left at 30° below the horizontal. Each chain fails at 600 lb. **Find** the maximum F that can be supported.

Sketch:



Free-body diagram of joint A. AB direction: $(\frac{3}{5}, \frac{4}{5})$.

Equations & Solution:

Joint A is a particle in equilibrium, so $\sum F_x = 0$ and $\sum F_y = 0$ (unknowns: T_{AB} , T_{AC} in terms of F).

$$\sum F_y = 0 : T_{AB} \left(\frac{4}{5}\right) - F \sin 30^\circ = 0 \Rightarrow T_{AB} = \frac{0.5}{0.8} F = 0.625 F$$

$$\sum F_x = 0 : T_{AB} \left(\frac{3}{5}\right) + T_{AC} - F \cos 30^\circ = 0$$

$$T_{AC} = 0.866 F - 0.6(0.625 F) = 0.866 F - 0.375 F = 0.491 F$$

Check each chain against its limit ($T_{\max} = 600 \text{ lb}$):

$$T_{AB} = 600 \text{ lb} \Rightarrow F = \frac{600 \text{ lb}}{0.625} = 960 \text{ lb}$$

$$T_{AC} = 600 \text{ lb} \Rightarrow F = \frac{600 \text{ lb}}{0.491} = 1220 \text{ lb}$$

The smaller F governs: chain AB fails first. At $F = 960 \text{ lb}$, $T_{AC} = 0.491(960 \text{ lb}) = 471 \text{ lb} < 600 \text{ lb}$, so AC is safe.

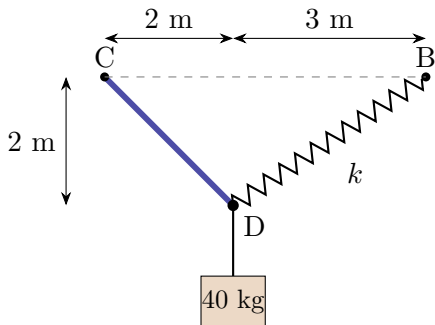
Answer

$F_{\max} = 960 \text{ lb}$ (chain AB reaches 600 lb first; $T_{AC} = 471 \text{ lb}$ at that load).

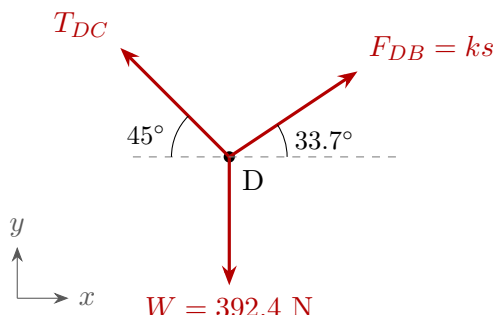
Problem 5

Given / Find: A 40 kg crate hangs from ring D, held by cord DC and spring DB ($k = 180 \text{ N/m}$). C is 2 m left and 2 m above D; B is 3 m right and 2 m above D. **Find** the unstretched length ℓ_0 of the spring DB.

Sketch:



Setup



FBD of ring D

Equations & Solution:

Weight and geometry.

$$W = mg = (40 \text{ kg})(9.81 \text{ m/s}^2) = 392.4 \text{ N}$$

$$L_{DC} = \sqrt{2^2 + 2^2} = \sqrt{8} \text{ m}, \quad L_{DB} = \sqrt{3^2 + 2^2} = \sqrt{13} = 3.606 \text{ m}$$

Equilibrium of ring D (the vertical cord DA carries the full weight W):

$$\sum F_x = 0: \quad -T_{DC} \frac{2}{\sqrt{8}} + F_{DB} \frac{3}{\sqrt{13}} = 0 \Rightarrow T_{DC} = \frac{3\sqrt{2}}{\sqrt{13}} F_{DB}$$

$$\sum F_y = 0: \quad T_{DC} \frac{2}{\sqrt{8}} + F_{DB} \frac{2}{\sqrt{13}} - W = 0$$

Substituting $T_{DC} (2/\sqrt{8}) = T_{DC}/\sqrt{2} = 3F_{DB}/\sqrt{13}$:

$$\frac{3F_{DB}}{\sqrt{13}} + \frac{2F_{DB}}{\sqrt{13}} = W \Rightarrow F_{DB} = \frac{\sqrt{13}}{5} W = \frac{3.606}{5} (392.4 \text{ N}) = 283 \text{ N}$$

(and $T_{DC} = 333 \text{ N}$).

Spring law $F_{DB} = ks$ gives the stretch, and $s = L_{DB} - \ell_0$ gives the unstretched length:

$$s = \frac{F_{DB}}{k} = \frac{283 \text{ N}}{180 \text{ N/m}} = 1.57 \text{ m} \quad \Rightarrow \quad \ell_0 = L_{DB} - s = 3.606 \text{ m} - 1.572 \text{ m} = 2.03 \text{ m}$$

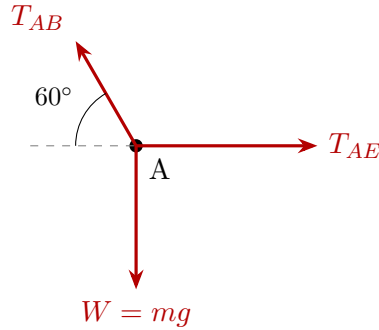
Answer

$\ell_0 = 2.03 \text{ m}$ (spring force $F_{DB} = 283 \text{ N}$, stretch $s = 1.57 \text{ m}$).

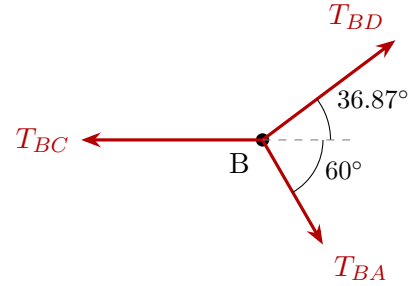
Problem 6

Given / Find: Cords BC, BD, AB, AE and AH hold a hook H at A. Cord BD has a 3-4-5 slope (4 horizontal, 3 vertical), AB makes 60° with the horizontal, BC and AE are horizontal. Each cord carries at most 500 N. **Find** the largest mass m that can hang from H.

Sketch:



FBD of joint A



FBD of joint B

Equations & Solution:

Cord AB pulls A up and to the left, so at B the cord pulls back down and to the right ($T_{BA} = T_{AB}$). Work in terms of the weight $W = mg$, then apply the 500 N limit to the most heavily loaded cord.

Joint A ($T_{AH} = W$ supports the hook):

$$\begin{aligned}\sum F_y = 0 : \quad T_{AB} \sin 60^\circ - W &= 0 \Rightarrow T_{AB} = \frac{W}{\sin 60^\circ} = 1.155 W \\ \sum F_x = 0 : \quad T_{AE} - T_{AB} \cos 60^\circ &= 0 \Rightarrow T_{AE} = 0.577 W\end{aligned}$$

Joint B ($T_{BA} = T_{AB} = 1.155W$, direction $(\cos 60^\circ, -\sin 60^\circ)$; BD direction $(\frac{4}{5}, \frac{3}{5})$):

$$\begin{aligned}\sum F_y = 0 : \quad \frac{3}{5}T_{BD} - T_{AB} \sin 60^\circ &= 0 \Rightarrow T_{BD} = \frac{W}{0.6} = 1.667 W \\ \sum F_x = 0 : \quad \frac{4}{5}T_{BD} + T_{AB} \cos 60^\circ - T_{BC} &= 0 \Rightarrow T_{BC} = 0.8(1.667 W) + 0.5(1.155 W) = 1.911 W\end{aligned}$$

The largest tension coefficient is $T_{BC} = 1.911 W$:

$$T_{BC} = 500 \text{ N} \Rightarrow W = \frac{500 \text{ N}}{1.911} = 261.7 \text{ N} \Rightarrow m = \frac{W}{g} = \frac{261.7 \text{ N}}{9.81 \text{ m/s}^2} = 26.7 \text{ kg}$$

Check the others at this load: $T_{BD} = 436 \text{ N}$, $T_{AB} = 302 \text{ N}$, $T_{AE} = 151 \text{ N}$, all below 500 N.

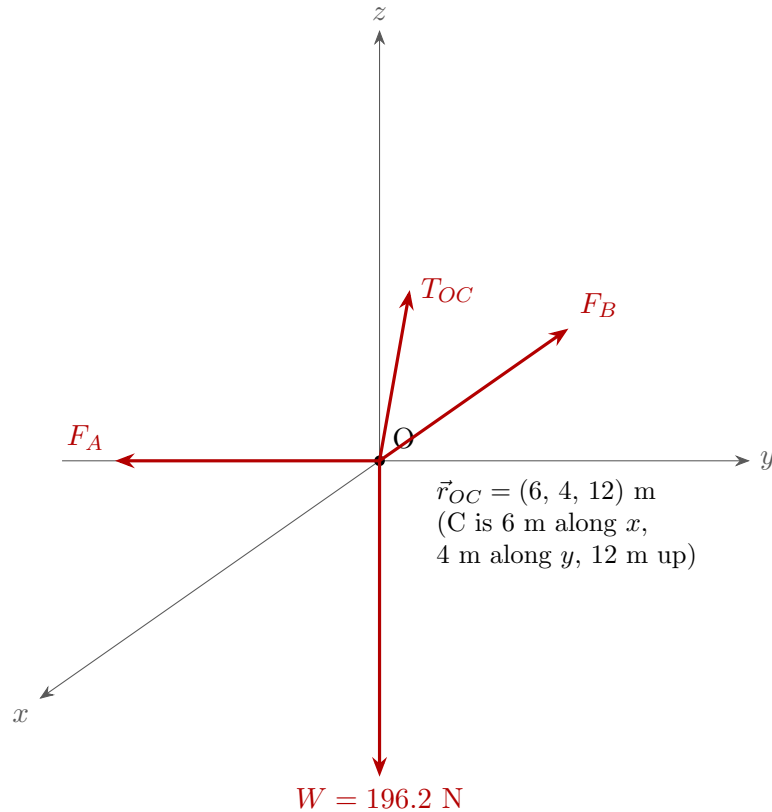
Answer

$m_{\max} = 26.7 \text{ kg}$ (cord BC governs, at $T_{BC} = 500 \text{ N}$).

Problem 7

Given / Find: A 20 kg crate hangs from O, held by two springs OA and OB (each $\ell_0 = 2$ m, $k = 300$ N/m) lying along the horizontal axes and cord OC whose end C is 6 m and 4 m horizontally and 12 m vertically from O. **Find** the stretch s_{OA} and s_{OB} in each spring for equilibrium.

Sketch:



FBD of O: F_A pulls toward A (along $-y$), F_B toward B (along $-x$); $\vec{r}_{OC} = (6, 4, 12)$ m.

Equations & Solution:

Weight and cord direction.

$$W = (20 \text{ kg})(9.81 \text{ m/s}^2) = 196.2 \text{ N}$$

$$\vec{r}_{OC} = (6\hat{i} + 4\hat{j} + 12\hat{k}) \text{ m}, \quad |\vec{r}_{OC}| = \sqrt{36 + 16 + 144} = 14 \text{ m}$$

$$\vec{T}_{OC} = T_{OC} \frac{6\hat{i} + 4\hat{j} + 12\hat{k}}{14}, \quad \vec{F}_B = -F_B \hat{i}, \quad \vec{F}_A = -F_A \hat{j}, \quad \vec{W} = -196.2 \hat{k} \text{ N}$$

Equilibrium of O, $\sum \vec{F} = \vec{0}$:

$$\sum F_z = 0 : \quad \frac{12}{14}T_{OC} - 196.2 \text{ N} = 0 \Rightarrow T_{OC} = 228.9 \text{ N}$$

$$\sum F_x = 0 : \quad \frac{6}{14}T_{OC} - F_B = 0 \Rightarrow F_B = \frac{6}{14}(228.9 \text{ N}) = 98.1 \text{ N}$$

$$\sum F_y = 0 : \quad \frac{4}{14}T_{OC} - F_A = 0 \Rightarrow F_A = \frac{4}{14}(228.9 \text{ N}) = 65.4 \text{ N}$$

Spring law $F = ks$ (the unstretched length $\ell_0 = 2 \text{ m}$ is not needed for the stretch):

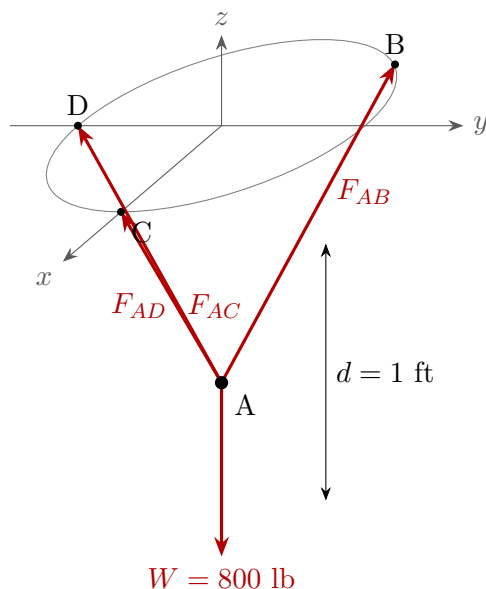
$$s_{OA} = \frac{F_A}{k} = \frac{65.4 \text{ N}}{300 \text{ N/m}} = 0.218 \text{ m}, \quad s_{OB} = \frac{F_B}{k} = \frac{98.1 \text{ N}}{300 \text{ N/m}} = 0.327 \text{ m}$$

Answer
$s_{OA} = 0.218 \text{ m}$ and $s_{OB} = 0.327 \text{ m}$ (spring forces 65.4 N and 98.1 N; cord tension $T_{OC} = 229 \text{ N}$).

Problem 8

Given / Find: An 800 lb cylinder hangs from ring A, supported by chains AB, AC and AD attached to a horizontal circle of radius 1 ft (in the $z = 0$ plane), with $d = 1$ ft below it. C is on the $+x$ axis, D on the $-y$ axis, and B lies 135° from each of them. **Find** the tension in each chain.

Sketch:



FBD of ring A (all three chains pull A up toward B, C, D). Points: A(0, 0, -1), B(-0.707, 0.707, 0), C(1, 0, 0), D(0, -1, 0) ft.

Equations & Solution:

Chain vectors from A (A at (0, 0, -1) ft; the anchors lie on the unit circle, with B at 135° from both C and D):

$$\begin{aligned}\vec{r}_{AB} &= (-0.707, 0.707, 1) \text{ ft}, & |\vec{r}_{AB}| &= \sqrt{0.5 + 0.5 + 1} = 1.414 \text{ ft} \\ \vec{r}_{AC} &= (1, 0, 1) \text{ ft}, & |\vec{r}_{AC}| &= 1.414 \text{ ft} \\ \vec{r}_{AD} &= (0, -1, 1) \text{ ft}, & |\vec{r}_{AD}| &= 1.414 \text{ ft}\end{aligned}$$

Force vectors ($\vec{F} = F \vec{r}/|\vec{r}|$), plus the weight $\vec{W} = -800 \hat{k}$ lb:

$$\begin{aligned}\vec{F}_{AB} &= F_{AB} (-0.5 \hat{i} + 0.5 \hat{j} + 0.707 \hat{k}), \\ \vec{F}_{AC} &= F_{AC} (0.707 \hat{i} + 0.707 \hat{k}), & \vec{F}_{AD} &= F_{AD} (-0.707 \hat{j} + 0.707 \hat{k})\end{aligned}$$

Equilibrium of A, $\sum \vec{F} = \vec{0}$:

$$\begin{aligned}\sum F_x &= 0 : & -0.5 F_{AB} + 0.707 F_{AC} &= 0 \Rightarrow F_{AC} = 0.707 F_{AB} \\ \sum F_y &= 0 : & 0.5 F_{AB} - 0.707 F_{AD} &= 0 \Rightarrow F_{AD} = 0.707 F_{AB} \\ \sum F_z &= 0 : & 0.707 (F_{AB} + F_{AC} + F_{AD}) - 800 \text{ lb} &= 0\end{aligned}$$

Substituting the first two into the third: $0.707 F_{AB}(1+0.707+0.707) = 800 \text{ lb} \Rightarrow 1.707 F_{AB} = 800 \text{ lb}$,
so

$$F_{AB} = \frac{800 \text{ lb}}{1.707} = 469 \text{ lb}, \quad F_{AC} = F_{AD} = 0.707 (468.6 \text{ lb}) = 331 \text{ lb}$$

Check: $0.707(469 + 331 + 331) \text{ lb} = 800 \text{ lb} \checkmark$

Answer
$F_{AB} = 469 \text{ lb}, F_{AC} = 331 \text{ lb}, F_{AD} = 331 \text{ lb}.$